Telco Customer Churn Analysis – Project Summary

This project performs **Exploratory Data Analysis (EDA)** on a telecom customer churn dataset to identify the key factors influencing customer attrition.

1. Data Cleaning & Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset from Excel using **pandas**
- Checked missing values and data types

This ensured the dataset was clean and ready for analysis.

2. Overall Churn Distribution

- Count plot and Pie Chart were created.
- The dataset shows a noticeable portion of customers have churned.
- Majority of customers remain active, but churn percentage is significant enough to require business attention.

3. Demographic Analysis

Gender

- Churn distribution is almost similar across genders.
- Gender does not strongly influence churn.

Senior Citizen

- Senior citizens show comparatively higher churn percentage.
- Age category plays a moderate role in churn behavior.

5. Payment Method Analysis

- Certain payment methods show higher churn counts.
- Customers using specific billing methods (e.g., electronic check in typical datasets) tend to churn more.
- Payment behavior is an important churn indicator.

4. Tenure Analysis

Tenure Histogram

- Customers with lower tenure (new customers) show higher churn.
- Long-term customers are more stable.

Tenure Groups (Contract-Based)

- Month-to-month customers churn more.
- One-year customers show moderate churn.
- Two-year contract customers are more stable.

Insight: Longer commitment reduces churn probability.

6. Service-Based Analysis (Major Insights)

The following services were analyzed:

- PhoneService
- MultipleLines
- InternetServices
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies

Overall Conclusion

Customer churn is mainly influenced by:

- Short tenure (new customers)
- Month-to-month contracts
- Lack of security and support services
- Certain payment methods
- Senior citizen category (moderate effect)

Business Recommendation:

To reduce churn, telecom companies should:

- Encourage long-term contracts
- Promote bundled services (security + tech support)
- Target high-risk customers (new & month-to-month)
- Improve customer engagement during early tenure